

● RAG Learning

3.3 Compound Measures & Rates of Change

Pearson Edexcel GCSE Mathematics 1MA1 — Ratio, proportion and rates of change — spec refs R1, R11, R14, R15 — Papers 1 (non-calculator), 2 and 3 (calculator)

3.3.1 Converting Between Standard Units (R1)

Every compound measure question starts with a units check. If a speed is in km/h, the time must be in hours; if a pressure is wanted in N/m², the area must be in m². Edexcel mark schemes usually give the method mark only when the conversion has been done correctly, so convert **first**, then calculate.

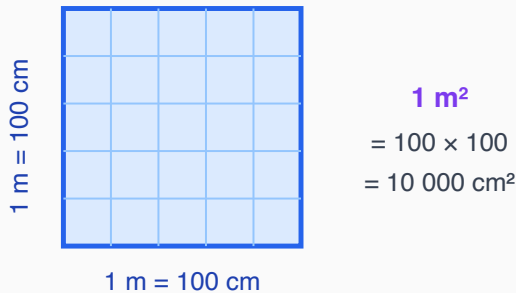
Length, mass, capacity and time

Quantity	Facts you must know
Length	10 mm = 1 cm 100 cm = 1 m 1000 m = 1 km
Mass	1000 mg = 1 g 1000 g = 1 kg 1000 kg = 1 tonne
Capacity	1000 ml = 1 litre 100 cl = 1 litre 1 ml = 1 cm ³ 1000 litres = 1 m ³
Time	60 s = 1 min 60 min = 1 h 3600 s = 1 h 24 h = 1 day

Exam Tip: Hours are not decimal minutes. 1.4 hours is 1 hour 24 minutes ($0.4 \times 60 = 24$), not 1 hour 40 minutes. Going the other way, 50 minutes = $\frac{50}{60} = \frac{5}{6}$ hour — on Paper 1 keep it as a fraction rather than writing 0.83.

Area and volume units

A square centimetre is a square 10 mm by 10 mm, so it contains $10 \times 10 = 100 \text{ mm}^2$. Area conversions use the length factor **squared** and volume conversions use it **cubed**.



Area: $1 \text{ cm}^2 = 100 \text{ mm}^2$ $1 \text{ m}^2 = 10\,000 \text{ cm}^2$ $1 \text{ km}^2 = 1\,000\,000 \text{ m}^2$

Volume: $1 \text{ cm}^3 = 1000 \text{ mm}^3$ $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$

Length factor $k \rightarrow$ area factor $k^2 \rightarrow$ volume factor k^3 .

Worked Example: Change 7200 cm^3 into m^3 .

$1 \text{ m}^3 = 100^3 = 1\,000\,000 \text{ cm}^3$, so divide:

$$7200 \div 1\,000\,000 = \mathbf{0.0072 \text{ m}^3}$$

Worked Example: Change 0.65 m^2 into cm^2 .

$$0.65 \times 10\,000 = \mathbf{6500 \text{ cm}^2}$$

3.3.2 Compound Units (R1, R11)

A **compound unit** is made from two units and describes a rate: how much of one quantity there is for each unit of another. Examples: m/s, km/h, mph, £ per hour, pence per gram, g/cm^3 , kg/m^3 , N/m^2 .

The "/" or the word "per" means **divide**. A unit price in pence per gram is $\text{pence} \div \text{grams}$; a density in g/cm^3 is $\text{grams} \div \text{cm}^3$. If you forget a formula, the unit tells you what to divide by what.

Converting compound units

Convert the top unit and the bottom unit separately, then divide.

Worked Example: Change 63 km/h into m/s.

$$63 \text{ km} = 63\,000 \text{ m} \quad 1 \text{ hour} = 3600 \text{ s}$$

$$63\,000 \div 3600 = \mathbf{17.5 \text{ m/s}}$$

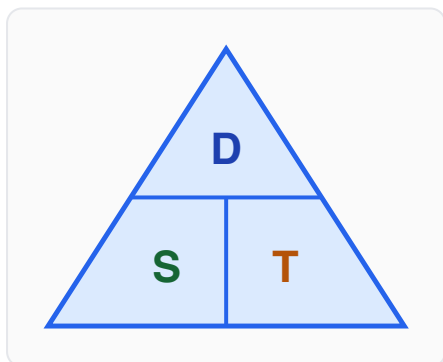
$$\text{Check: km/h} \div 3.6 = \text{m/s, and } 63 \div 3.6 = 17.5 \checkmark$$

Worked Example: The density of copper is 8.9 g/cm³. Write this in kg/m³.

$$1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3, \text{ so } 1 \text{ m}^3 \text{ of copper has mass } 8.9 \times 1\,000\,000 = 8\,900\,000 \text{ g} = 8900 \text{ kg}$$

$$\text{Density} = \mathbf{8900 \text{ kg/m}^3} \quad (\text{g/cm}^3 \rightarrow \text{kg/m}^3: \text{multiply by } 1000)$$

3.3.3 Speed, Distance and Time (R11)



$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Distance} = \text{Speed} \times \text{Time} \quad \text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

Cover the letter you want: what is left shows the calculation.

Worked Example: A taxi travels for 35 minutes at an average speed of 84 km/h. How far does it travel?

$$35 \text{ minutes} = \frac{35}{60} \text{ hour}$$

$$\text{Distance} = 84 \times \frac{35}{60} = \mathbf{49 \text{ km}}$$

Worked Example: A van travels 117 miles at an average speed of 52 mph. It sets off at 13:50. At what time does it arrive?

$$\text{Time} = 117 \div 52 = 2.25 \text{ hours} = 2 \text{ hours } 15 \text{ minutes}$$

$$\text{Arrival} = 13:50 + 2 \text{ h } 15 \text{ min} = \mathbf{16:05}$$

Average speed over several stages

Average speed = **total distance** $\div$ **total time**. Never average the speeds themselves — the slower stage takes longer, so it counts for more.

Worked Example: A motorbike travels 45 km at 90 km/h and then another 45 km at 60 km/h. Work out its average speed.

$$\text{Time 1} = 45 \div 90 = 0.5 \text{ h} \quad \text{Time 2} = 45 \div 60 = 0.75 \text{ h}$$

$$\text{Average speed} = \frac{90}{1.25} = \mathbf{72 \text{ km/h}} \quad (\text{not } 75 \text{ km/h})$$

Exam Tip: If the question says “for the whole journey”, include any time spent stopped. Always give units, and sense-check: walking ≈ 5 km/h, cycling ≈ 20 km/h, motorway driving ≈ 110 km/h.

3.3.4 Rates of Pay and Unit Pricing (R11)

Edexcel often sets “best buy” and “who is paid more?” problems, typically worth 3 or 4 marks. The method is always the same: reduce every option to the **same rate**, then compare and write a conclusion.

Worked Example: Jar A: 340 g of jam for £2.04. Jar B: 454 g of jam for £2.59. Which jar is better value?

$$\text{Jar A: } 204 \div 340 = 0.6\text{p per gram}$$

Jar B: $259 \div 454 = 0.570 \dots$ p per gram

Jar B is better value, because it costs less per gram ($0.57\text{p} < 0.6\text{p}$).

Worked Example: Dev is paid £12.20 per hour. Erin is paid £24 960 a year for working 40 hours a week, 52 weeks a year. Who has the higher hourly rate?

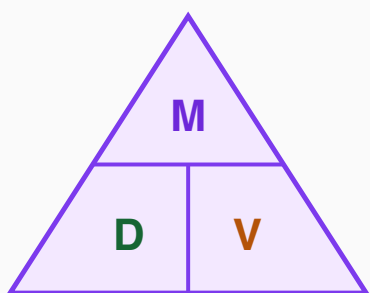
Erin's hours = $40 \times 52 = 2080$ Erin's rate = $24\,960 \div 2080 = \text{£}12.00$ per hour

Dev earns 20p per hour more.

Exam Tip: Edexcel awards the final mark (C1) for a conclusion supported by correct figures. Writing "Jar B" with no comparison, or comparing price per gram for one jar with grams per penny for the other, will lose marks.

3.3.5 Density (R11)

Density is mass per unit volume. It is usually measured in g/cm^3 or kg/m^3 . Learn the formula and its rearrangements — do not rely on it being given in the question.



$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$\text{Mass} = \text{Density} \times \text{Volume} \quad \text{Volume} = \frac{\text{Mass}}{\text{Density}}$$

Worked Example: A brass cuboid measures 8 cm by 5 cm by 3 cm and has a mass of 1020 g. Work out the density of brass.

$$\text{Volume} = 8 \times 5 \times 3 = 120 \text{ cm}^3$$

$$\text{Density} = 1020 \div 120 = \mathbf{8.5 \text{ g/cm}^3}$$

Worked Example: 400 cm³ of liquid P (density 1.1 g/cm³) is mixed with 100 cm³ of liquid Q (density 0.8 g/cm³). Work out the density of the mixture.

$$\text{Mass of P} = 400 \times 1.1 = 440 \text{ g} \quad \text{Mass of Q} = 100 \times 0.8 = 80 \text{ g}$$

$$\text{Density} = \frac{440 + 80}{400 + 100} = \frac{520}{500} = \mathbf{1.04 \text{ g/cm}^3}$$

Exam Tip: For a mixture or alloy, work with **totals**: total mass $\div$ total volume. The mean of the two densities (0.95 here) is wrong unless the volumes happen to be equal.

3.3.6 Pressure (R11)

Pressure is force per unit area. Force is measured in newtons (N), so pressure is in N/m² (pascals) or N/cm². The formula $P = \frac{F}{A}$ is sometimes printed in the question, but you should know it.

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} \quad \text{Force} = \text{Pressure} \times \text{Area} \quad \text{Area} = \frac{\text{Force}}{\text{Pressure}}$$

The same force spread over a **larger** area gives a **smaller** pressure. A cuboid exerts its greatest pressure when it stands on its smallest face.

Worked Example: A crate weighs 150 N. Its base is a rectangle 50 cm by 60 cm. Work out the pressure on the ground in N/m².

$$\text{Area} = 0.5 \times 0.6 = 0.3 \text{ m}^2$$

$$\text{Pressure} = 150 \div 0.3 = \mathbf{500 \text{ N/m}^2}$$

Worked Example: A cuboid 25 cm × 12 cm × 4 cm has a weight of 60 N. Work out the greatest pressure it can exert, in N/cm².

$$\text{Smallest face} = 12 \times 4 = 48 \text{ cm}^2$$

$$\text{Greatest pressure} = 60 \div 48 = \mathbf{1.25 \text{ N/cm}^2}$$

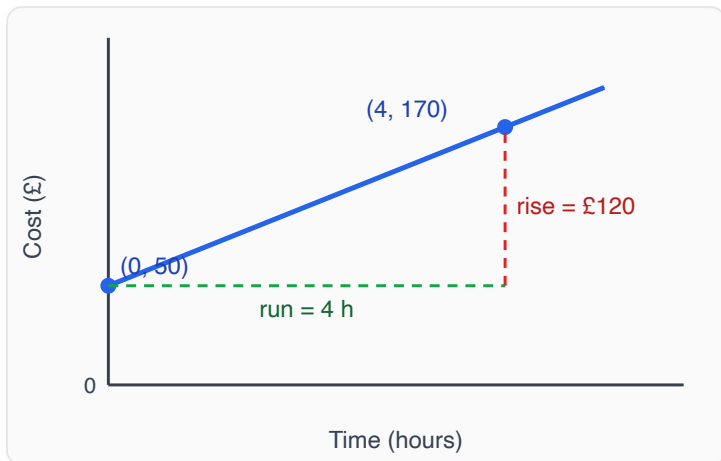
Exam Tip: Convert lengths to metres *before* working out an area in m². 50 cm × 60 cm = 3000 cm², and 3000 cm² = 0.3 m² (divide by 10 000, not by 100).

3.3.7 Gradient of a Straight Line as a Rate of Change (R14)

On a straight-line graph the gradient is constant, so the rate of change is the same everywhere. The units of the gradient are the *y*-axis units **per** *x*-axis unit.

$$\text{Gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Graph	Gradient means	Units
Distance–time	Speed	m/s, km/h
Velocity–time	Acceleration	m/s ²
Cost–time (e.g. electrician)	Charge per hour	£ per hour
Depth of water–time	Rate of filling or emptying	cm per minute
Conversion graph (e.g. litres to gallons)	Conversion rate	gallons per litre



Worked Example: An electrician's charges are shown by a straight line through (0, 50) and (4, 170).

$$\text{Gradient} = \frac{170 - 50}{4 - 0} = \frac{120}{4} = 30 \rightarrow \text{the electrician charges } \mathbf{\pounds 30 \text{ per hour}}.$$

The y -intercept, 50, is a fixed **call-out charge of £50**.

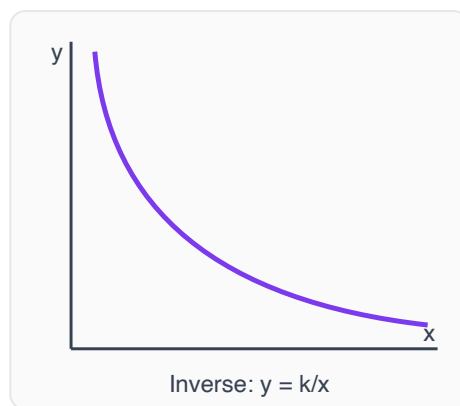
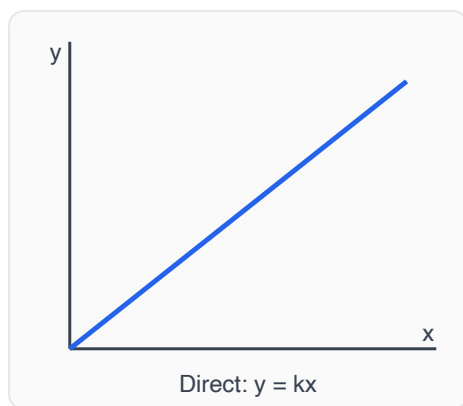
$$\text{Cost of a 6-hour job: } 30 \times 6 + 50 = \mathbf{\pounds 230}$$

Worked Example: The depth of water in a pond falls steadily from 96 cm at $t = 0$ to 36 cm at $t = 24$ minutes.

$$\text{Gradient} = \frac{36 - 96}{24} = -2.5 \rightarrow \text{the depth } \mathbf{\text{falls by 2.5 cm every minute}}.$$

Exam Tip: "Interpret the gradient" means describe it in context with units: "the electrician charges £30 for each hour worked". A bare number such as "30" does not usually get the mark.

3.3.8 Graphs of Direct and Inverse Proportion (R14)



- **Direct proportion** ($y = kx$): a straight line **through the origin**. Its gradient is the constant of proportionality k . If x is multiplied by 3, so is y .
- **Inverse proportion** ($y = \frac{k}{x}$): for $x > 0$, a curve that falls and approaches both axes without ever reaching them. If x is multiplied by 3, y is divided by 3, and the product $xy = k$ stays the same.

Worked Example: A graph of the number of workers, w , against the time, t days, to build a wall is a curve through (2, 12) and (6, 4). Is this inverse proportion?

$2 \times 12 = 24$ and $6 \times 4 = 24$. The product is constant, so it is consistent with **inverse proportion**, $t = \frac{24}{w}$.

Exam Tip: A straight line that does **not** go through the origin (like the electrician's charges) is linear but is **not** direct proportion: doubling the hours does not double the cost.

3.3.9 Average and Instantaneous Rates of Change (R15) Higher

On a **curve** the gradient keeps changing, so there are two different rates to consider. R15 is Higher tier only, and the 1MA1 specification states it does not involve calculus — you find these gradients from chords and tangents.

Chord: a straight line joining two points on a curve. Its gradient is the **average rate of change** between those two points.

Tangent: a straight line that just touches the curve at one point. Its gradient is the **instantaneous rate of change** at that point (the gradient of the curve there).



Average rate of change from $x = a$ to $x = b$ (gradient of the chord):

$$\frac{y(b) - y(a)}{b - a}$$

Instantaneous rate of change at $x = a$: draw the tangent at $x = a$, read two well-separated points on it, and work out rise $\div$ run.

Worked Example: Work out the average rate of change of $y = x^2 + 3x$ between $x = 1$ and $x = 3$.

$$y(1) = 1 + 3 = 4 \quad y(3) = 9 + 9 = 18$$

$$\text{Gradient of chord} = \frac{18 - 4}{3 - 1} = \frac{14}{2} = 7$$

Estimating a gradient by drawing a tangent

1. Line up a ruler so that it touches the curve at the point, with equal gaps between ruler and curve on either side.

2. Draw the tangent long enough to read easily.
3. Choose two points on the tangent that are far apart (ideally on grid lines).
4. Gradient = change in $y \div$ change in x . State what it means, with units.

Worked Example: A distance–time graph for a runner is a curve. The tangent drawn at $t = 4$ s passes through (1, 2) and (7, 50), with distance in metres.

$$\text{Gradient} = \frac{50 - 2}{7 - 1} = \frac{48}{6} = 8$$

The runner's speed at $t = 4$ s is approximately **8 m/s**.

Worked Example: The volume of water in a tank after t minutes is $V = 3t^2$ litres.

Average rate from $t = 1$ to $t = 4$: $\frac{48 - 3}{4 - 1} = \frac{45}{3} = 15$ litres per minute.

The tangent at $t = 2$ passes through (1, 0) and (3, 24): gradient = $\frac{24 - 0}{3 - 1} = 12$.

So water is flowing in at about **12 litres per minute** at the instant $t = 2$.

Interpreting the gradient of a curve

Curved graph	Gradient of chord	Gradient of tangent
Distance–time	Average speed over a time interval	Speed at an instant
Velocity–time	Average acceleration	Acceleration at an instant
Volume–time	Average rate of flow	Rate of flow at an instant
Population–time	Average growth rate	Growth rate at an instant

A distance–time curve that gets **steeper** shows an object **speeding up**; one that gets **flatter** shows it **slowing down**. Where the tangent is horizontal, the rate of change is zero at that instant — for example, a ball at the top of its flight.

Exam Tip: Edexcel accepts a range of answers for a gradient read from a drawn tangent, but your tangent must visibly touch the curve at the right point and your working (rise $\div$ run) must be shown. Say clearly whether you have found an average rate (chord) or an instantaneous rate (tangent).

Summary

Measure	Formula	Typical units
Speed	distance $\div$ time	m/s, km/h, mph
Density	mass $\div$ volume	g/cm ³ , kg/m ³
Pressure	force $\div$ area	N/m ² (Pa), N/cm ²
Rate of pay	pay $\div$ hours	£ per hour
Unit price	cost $\div$ quantity	p per g, £ per litre
Rate of change (straight line)	gradient = rise $\div$ run	<i>y</i> -units per <i>x</i> -unit
Rate of change (curve) Higher	chord = average; tangent = instantaneous	<i>y</i> -units per <i>x</i> -unit